

Viewpoint

Which direction in space? ☆

John M. Logsdon*

Elliott School of International Affairs, Space Policy Institute, George Washington University, Washington, DC 20052, USA

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Abstract

Two competing visions of future space activities have recently been articulated—one utilitarian, aiming to use space to make Earth a better place, the other more exploratory, aiming to expand the human and robotic presence in space. It is unlikely that resources will be available to pursue both, meaning that carefully considered choices will have to be made. For the first vision to work end users will have to be convinced that space can provide unique benefits reliably. For the second, a full political debate that reflects public opinion is required. It is essential that advocates of these different visions nevertheless coordinate their activities to avoid duplication, wasted resources and missed opportunities for synergy. An international discussion forum would be useful in this regard.

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During the next few years, spacefaring countries will be making decisions that will influence the direction of space activities for decades to come. As they do so, they have before them two very different visions of the character of future government activities in space. At issue is striking a productive balance between these two visions.

One vision looks inward, asking how activities in space can best serve the demands of Earth-bound society by delivering a variety of direct and tangible benefits to this planet's peoples. The other aims outward, driven by human curiosity, and focused on the exploration of the solar system and the universe beyond.

It is unlikely that public resources to pursue both visions vigorously will be made available by leading governments, at least in the short to mid-term. There are too many competing priorities for scarce public resources, and the case for investing more than is currently being spent on space activities is less than

totally compelling. Thus choices will be needed; they may well be different for different countries, leading to a 21st century array of space activities that is diverse, and potentially uncoordinated. If the latter turns out to be the case, there is the likelihood of duplication, wasted resources, and missed opportunities for synergy. It would be desirable if spacefaring countries could create some sort of forum within which they could discuss ways of avoiding these negative outcomes as they choose their future priorities in space.

1. Space in the service of society

The first of these two visions is most clearly laid out in the European Commission November 2003 'White Paper' *Space: A New European Frontier for An Expanding Union*.¹ While put forward in the European context, the analysis and conclusions of this document have global applicability. The White Paper argues that space technologies and their applications can make valuable contributions to society in areas such as economic growth, job creation and industrial competitiveness; sustainable development; security and defense; and in

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*Tel.: +1 202 994 7292; fax: +1 202 994 1639.

E-mail address: logsdon@gwu.edu.

¹European Commission COM (2003) 673.

fighting poverty and aiding the development of poorer societies. It suggests that the pursuit of these objectives should be the governing priority in selecting which areas of space capability development to pursue, and that such choices should be made by the potential users of space capabilities, not by those interested in developing them.

This perspective poses a challenge to the space community. That community must convince potential users that space capabilities should be the preferred means for achieving specific broader objectives; in many if not most cases, terrestrial alternatives exist. When users decide to invest in space development, the space sector must deliver performance as promised (see [1] for a discussion of this concept). Too often, space advocates have oversold the payoffs from the development of new space capabilities in order to justify the funding needed to bring them into being, only to produce disappointing results when those capabilities are put into operation. In the past, the space sector has been given special treatment by governments on the basis of its symbolic and strategic character. Once the space sector competes with other providers of similar capabilities, it can no longer expect such special treatment, which in the past has tended to ignore or discount the gap between its promises and its performance.

It is hard to argue against this utilitarian view of space. Because space capabilities do provide essential services in areas such as information transfer, Earth observation, positioning, timing and navigation, and support for terrestrial armed forces, there is no doubt that they will continue to be developed and used in the coming years. The question, though, is at what level of activity and thus funding. Are utilitarian purposes enough to continue to drive new developments in technology and lead to, for example, the substantial investments required to lower the costs of access to space? How many of these space applications qualify as public goods, thereby justifying government funding? Might they be provided on a profit-making basis by the private sector?

2. Exploring the unknown

Some believe that a utilitarian perspective should not dominate future space activities. For example, Daniel Goldin, former NASA Administrator, has recently argued that “a space program should never be about routine, guaranteed operations. If fear drives us to set only mediocre goals, then even meeting those goals is a form of failure”. Goldin suggests that “societal impact is strongest when we continue to stretch our necks and our imagination, searching for the loftiest of goals, not when we look down at the bottom line”. Rather than orient space activities toward Earth, “opening a new frontier—

going where no one has gone before—that is the ultimate impact of our space program” [2].

“Going where no one has gone before”—exploration—is the contending vision for the future of humanity’s 21st century space activities. Exploration includes science—the increase of knowledge—but it also has an inescapable geographical component, a sense of going somewhere on a mission of discovery. Robotic exploration of the solar system has been a long-standing goal of the US space program, and was an important activity of the space efforts of the USSR. In the past two decades, Japan and Europe have become active in space missions beyond the near vicinity of Earth. The recent landing of the Huygens spacecraft on the surface of Saturn’s moon Titan demonstrated Europe’s high level of robotic exploration capability. Other countries such as India and China are also planning robotic missions to the Moon.

Between December 1968 and December 1972, 27 American astronauts traveled to the moon; 12 of them walked on the lunar surface. These have been the first and last to date human missions of exploration, but the reason for undertaking them had little to do with an exploratory spirit. Rather, they were responses to an April 1961 directive to his advisers from President John F. Kennedy to identify “a space program which promises dramatic results in which we can win”(see [3] for an account of the origins of Project Apollo). Attempts in the USA since the Apollo program to gain approval for sending people to explore beyond Earth orbit have to date been unsuccessful.

On January 14, 2004 US President George W. Bush proposed what has come to be called a new “Vision for Space Exploration”, centered on “sustained and affordable human and robotic programs to explore the solar system and beyond” [4]. This new vision differs both from Project Apollo and earlier US proposals for human exploration by being planned on a “go as you can pay” basis. The first human mission back to the moon is scheduled for the 2015–2020 time frame, and the president set no deadline in the new exploration policy for sending people to Mars or other distant destinations. As President Bush said, this is to be “a journey, not a race”.² The US vision sets an outward direction for US space activities and provides a focus around which NASA can at least organize itself and its programs. The specifics of how to carry out the vision will constantly evolve.

In his January 14 speech, the president suggested that “like the explorers of the past”, it would not be possible to identify “all that we will gain from space exploration”, but “the eventual return will be great”. This is a very different rationale for space activities than that

²The President’s speech can be accessed at www.whitehouse.gov/news/releases/2004/01/20040114-3.html.

offered for the more utilitarian perspective. Whether it will resonate with the American public so that a commitment is indeed made to exploration as the primary objective of the government's civilian space activities is yet to be determined. In late November 2004, the US Congress provide full funding for the first year of the vision, but this was done as a last-minute, late-night maneuver, not on the basis of a full debate over the merits of the commitment that has been proposed. That political debate and the public sentiment it reflects will be a necessary prerequisite to the sustainability of the exploratory vision for the USA.

In his January 14 speech, President Bush also called "on other nations to join us on this journey, in a spirit of coopération and friendship". NASA convened a November 2004 workshop with potential international participants in implementing the vision. This workshop was attended by representatives of 17 other space agencies; all expressed interest in principle in becoming involved. Whether this interest can be converted into a decision to participate by the governments which oversee those space agencies is still an open question. NASA is moving quickly to define the initial steps in implementing the vision, and there may be a limited window of opportunity for international engagement in those steps. The terms and conditions for participation are likely to be different from those that have governed past international space cooperation, and that adds another dimension of complexity to making space exploration a globally shared vision. However, the US vision is intended to be a many-year undertaking, and thus there will be continuing opportunities for other countries and their scientists, astronauts and industries to participate in subsequent stages.

The choice between increased funding for utilitarian space activities and for space exploration is put in rather sharp perspective in the European Commission White Paper. That document lays out three funding scenarios for European space activities in the 2007–2013 period. Only in the most optimistic scenario, with funding growing by an average of 4.6% a year, is there substantial funding available for both space exploration and security space applications. If funding growth falls short of this optimistic rate, the White Paper suggests that priority should go to increased funding for security applications, with space exploration supported at a substantially lower level.³ Other non-European countries are likely to face a similar choice as they ponder how much of their space budget to spend on US-led exploratory activities, or indeed on their own exploratory efforts.

A focus on space exploration is not without its critics. The head of the small US space company AeroAstro, Rick Fleeter, has suggested that advocates of space

exploration reflect a false belief "that space's highest and true calling is achieving societal goals of research and exploration into the unknown". Fleeter suggests that this belief is "the burdensome baggage of an aristocratic calling, now bankrupt both ideologically and financially", which "is hindering our community from reaching our potential to serve humanity". To Fleeter, these "old ideas are rigid and anachronistic, no longer pointing to a brighter tomorrow, but rather back toward a dead end of technological progress for its own sake".⁴

Whatever the merits of this perspective, it states in colorful language the tension between a utilitarian and an exploratory vision for the future in space. In deciding how to strike a balance between these two visions, a number of questions must be addressed.

One set of questions has to do with the potential liabilities associated with using space systems to carry out centrally important functions for society. Such systems remain expensive to develop and launch. They have a mixed record of reliability, and repair of problems or failures is at best very difficult. Satellites and their associated ground systems are rather vulnerable to deliberate or unintentional interference. Can they be counted on to be available at all times, for the purposes for which they are intended, without requiring extreme attention? When these factors are taken into account, do space systems indeed compare favorably with terrestrial alternatives for carrying out the same function? Are there unique and valuable functions that only space systems can perform?

Another set of questions has to do with the lack of an immediately persuasive rationale for space exploration, and particularly for sending humans beyond Earth orbit. As President Bush has suggested, the benefits of exploration may not be evident for some time in the future, and thus must be taken on the basis of combining precedent and faith.

One reason for the development of the US Vision for Space Exploration was the criticism of the Columbia Accident Investigation Board in its August 2003 report of "the lack, over the past three decades, of any national mandate providing NASA a compelling mission requiring human presence in space" [5]. Is human exploration beyond Earth orbit such a "compelling mission"? Certainly the entry of China into the elite group of countries able on its own to send people into space, and the interest of countries such as India, Japan and some European states in developing the capability to do so, suggest that human space flight remains a desirable goal for leading countries. Is travel beyond Earth orbit the only sustainable rationale for human space flight? How widely shared is

³White Paper, p. 51.

⁴*Space News*, October 18, 2004.

the belief that human travel to other destinations in the solar system, and perhaps long duration stays and eventual settlement there, is a high priority societal goal?

Addressing these and other questions will be key to decisions on the future of government space programs around the world. If they are answered well, the 21st century could see the full realization of both the practical and inspirational potentials of space.

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